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Salahaddin University-Erbil
School of Science
Department of Information Technology (IT)

FIRST-YEAR UNDERGRADUATE · MATHEMATICS

COMPLEX NUMBERS

Properties of Complex Conjugate and Absolute Value



PRESENTED BY
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ACADEMIC YEAR
[ACADEMIC YEAR]

Why Complex Numbers?

?

Let's start with a simple question:

Can we find the square root of a negative number?

With real numbers only — no.

Squaring a real number never gives a negative result:

$2^2 = 4$	$(-2)^2 = 4$	$0^2 = 0$
positive	positive	zero

So an equation like $x^2 = -1$ has no real solution.

THE NEW IDEA

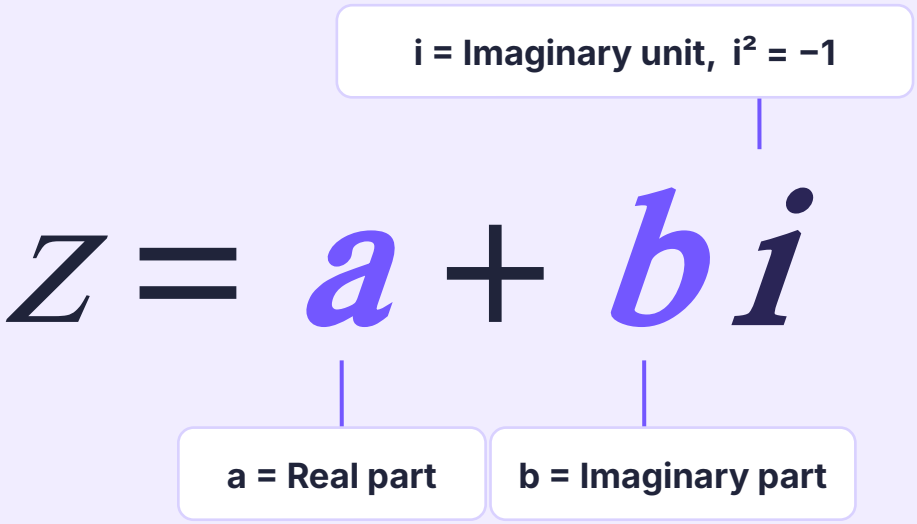
Mathematicians defined a new number, i, the imaginary unit:

$$i^2 = -1$$

$$i = \sqrt{(-1)}$$

Now $x^2 = -1$ has two answers:
$x = i$ and $x = -i$

What Is a Complex Number?



Here is an easy way to understand this: a complex number is simply a real number and an imaginary number added together.

TWO SIMPLE EXAMPLES

$z = 3 + 4i$

real part **3** imaginary part **4**

$z = 2 - 5i$

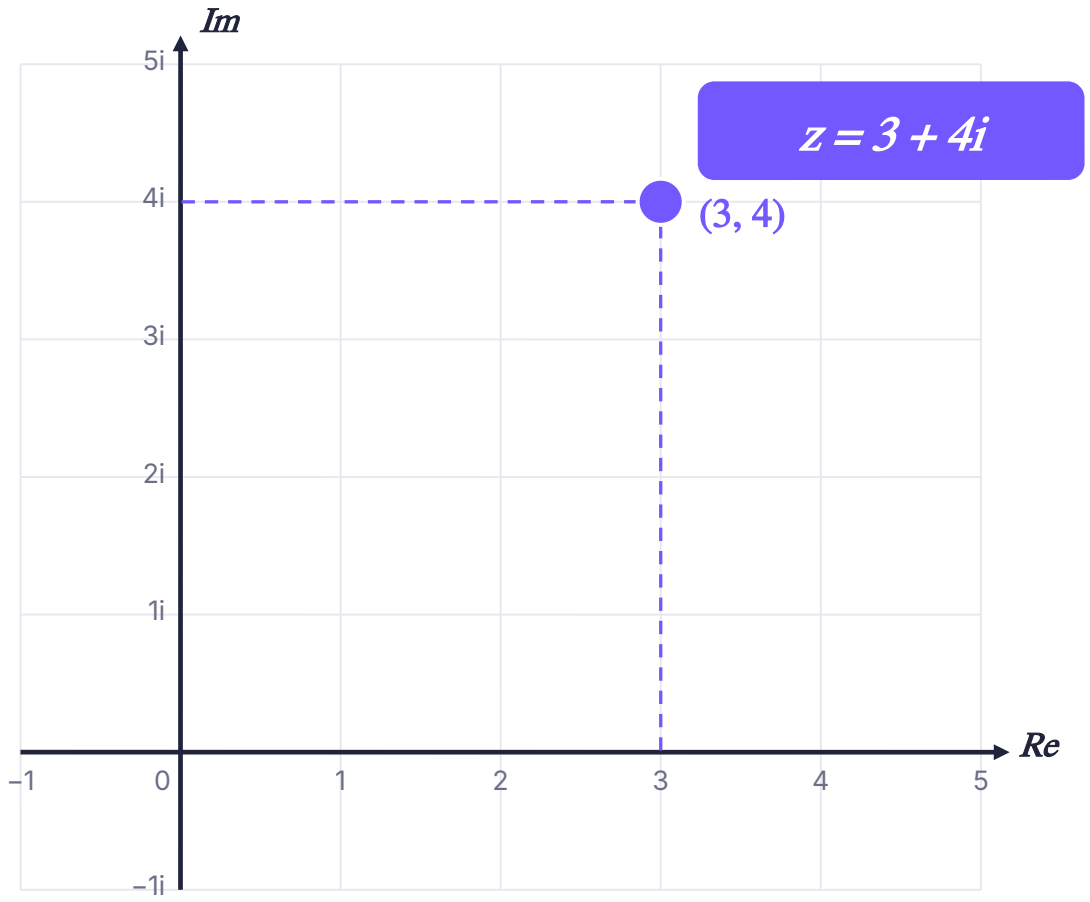
real part **2** imaginary part **−5**

Understanding Real and Imaginary Parts

Complex number z	Real part Re(z)	Imaginary part Im(z)
$z = 5 + 2i$	5	2
$z = -3 + 7i$	−3	7

Notice something interesting: the imaginary part is the number in front of i — it does **not** include i itself. For z = 5 + 2i, the imaginary part is 2, not 2i. Always keep the sign that comes with each number.

The Complex Plane



Horizontal axis = Real axis

It shows the real part a.

Vertical axis = Imaginary axis

It shows the imaginary part b.

Each complex number is a point

a + bi sits at the point (a, b).

To plot 3 + 4i: go 3 to the right, then 4 up.

What Is a Complex Conjugate?

DEFINITION

$$If \quad z = a + bi$$

$$then \quad \bar{z} = a - bi$$

$\bar{z}$ is read “z bar”.

Here is an easy way to remember it:

Keep the real part.
Flip the sign of the imaginary part.

Nothing else changes — only one sign.

EXAMPLE

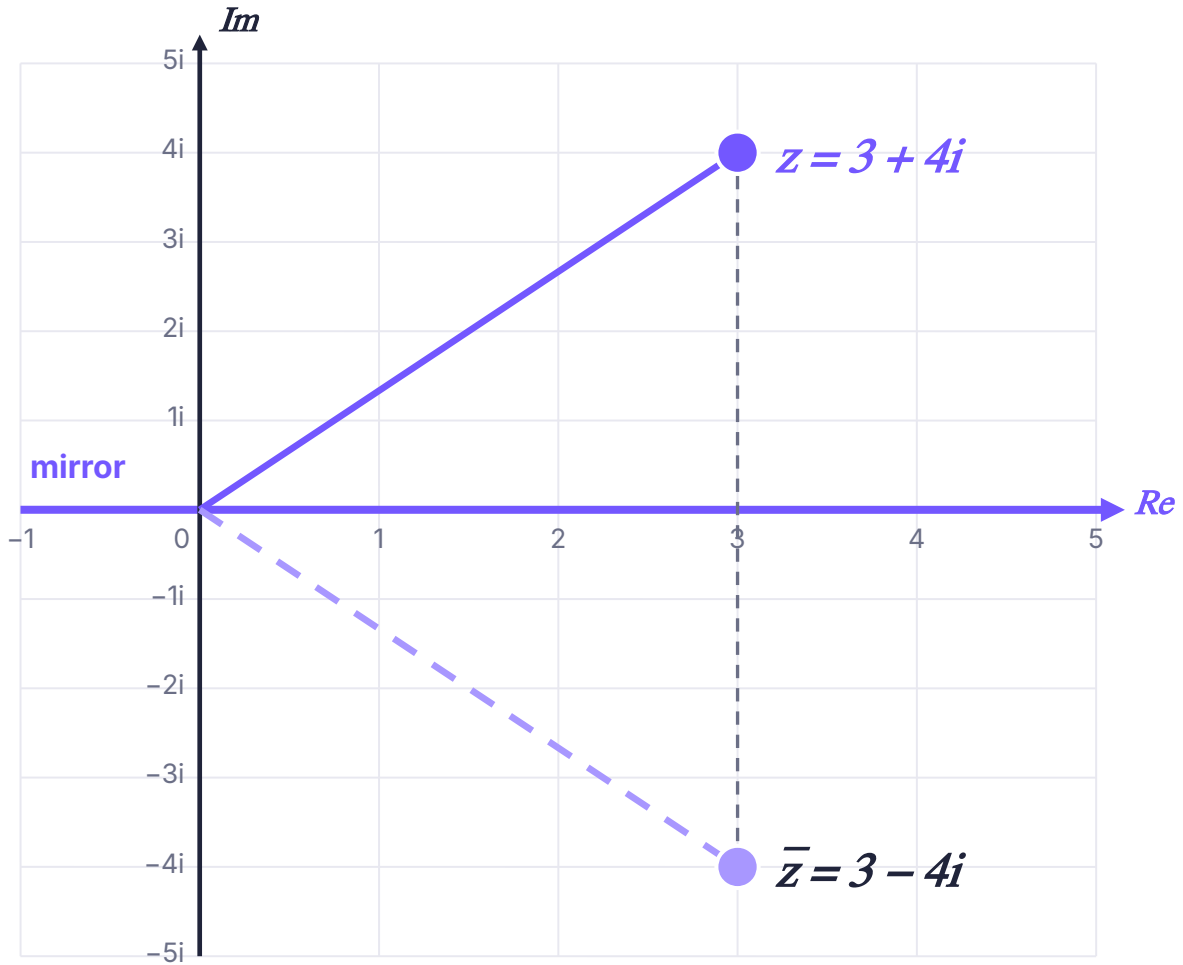
$$z = 3 + 4i$$

flip the sign

$$\bar{z} = 3 - 4i$$

Real part 3 stays the same.

Visualizing the Complex Conjugate



1 **Same distance to the right**

Both points have real part 3.

2 **Opposite vertical position**

4 units up becomes 4 units down.

3 **Real axis = the mirror**

The conjugate is a reflection across the real axis.

Fold the plane along the real axis: z lands on $\bar{z}$.

Basic Properties of Complex Conjugates

1 Conjugate of the conjugate

$$\overline{\overline{z}} = z$$

Flip the sign twice and you are back where you started.

EXAMPLE

$z = 2 + 3i$

$\overline{z} = 2 - 3i$

conjugate again: $2 + 3i$ ✓

2 Conjugate of a sum

$$\overline{(z_1 + z_2)} = \overline{z_1} + \overline{z_2}$$

Add then conjugate — or conjugate then add. Same answer.

EXAMPLE

$z_1 = 1 + 2i,\ z_2 = 3 + i$

$z_1 + z_2 = 4 + 3i \ \rightarrow\ 4 - 3i$

$(1 - 2i) + (3 - i) = 4 - 3i$ ✓

3 Conjugate of a product

$$\overline{(z_1 z_2)} = \overline{z_1}\ \overline{z_2}$$

Multiply then conjugate — or conjugate then multiply.

EXAMPLE

$z_1 = 1 + i,\ z_2 = 1 + 2i$

$z_1 z_2 = -1 + 3i \ \rightarrow\ -1 - 3i$

$(1 - i)(1 - 2i) = -1 - 3i$ ✓

Simple Examples of Complex Conjugates

EXAMPLE 1

$$z = 6 + 2i$$

$$\bar{z} = 6 - 2i$$

The + in front of 2i becomes −.

EXAMPLE 2

$$z = -4 - 3i$$

$$\bar{z} = -4 + 3i$$

The real part −4 keeps its sign. Only −3i becomes +3i.

EXAMPLE 3

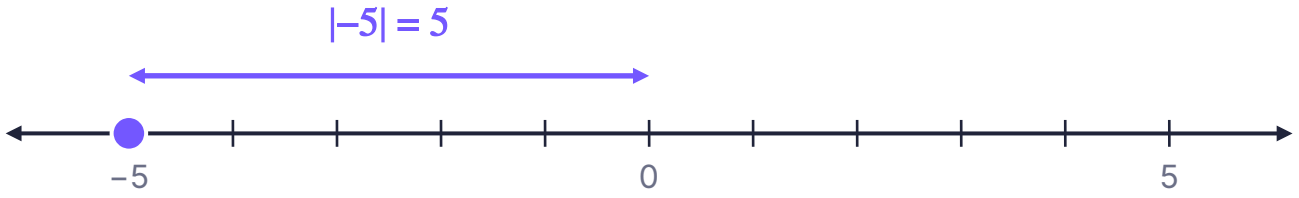
$$z = 7$$

$$\bar{z} = 7$$

There is no imaginary part (b = 0), so nothing flips. A real number is its own conjugate.

Introduction to Absolute Value

For real numbers, absolute value means distance from 0:



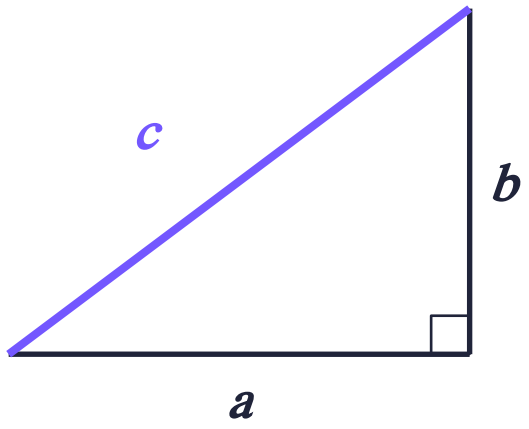
For a complex number, it means distance from the origin:

If $z = a + bi$,

$$|z| = \sqrt{(a^2 + b^2)}$$

It is also called the modulus or magnitude of z. Read |z| as “the absolute value of z”.

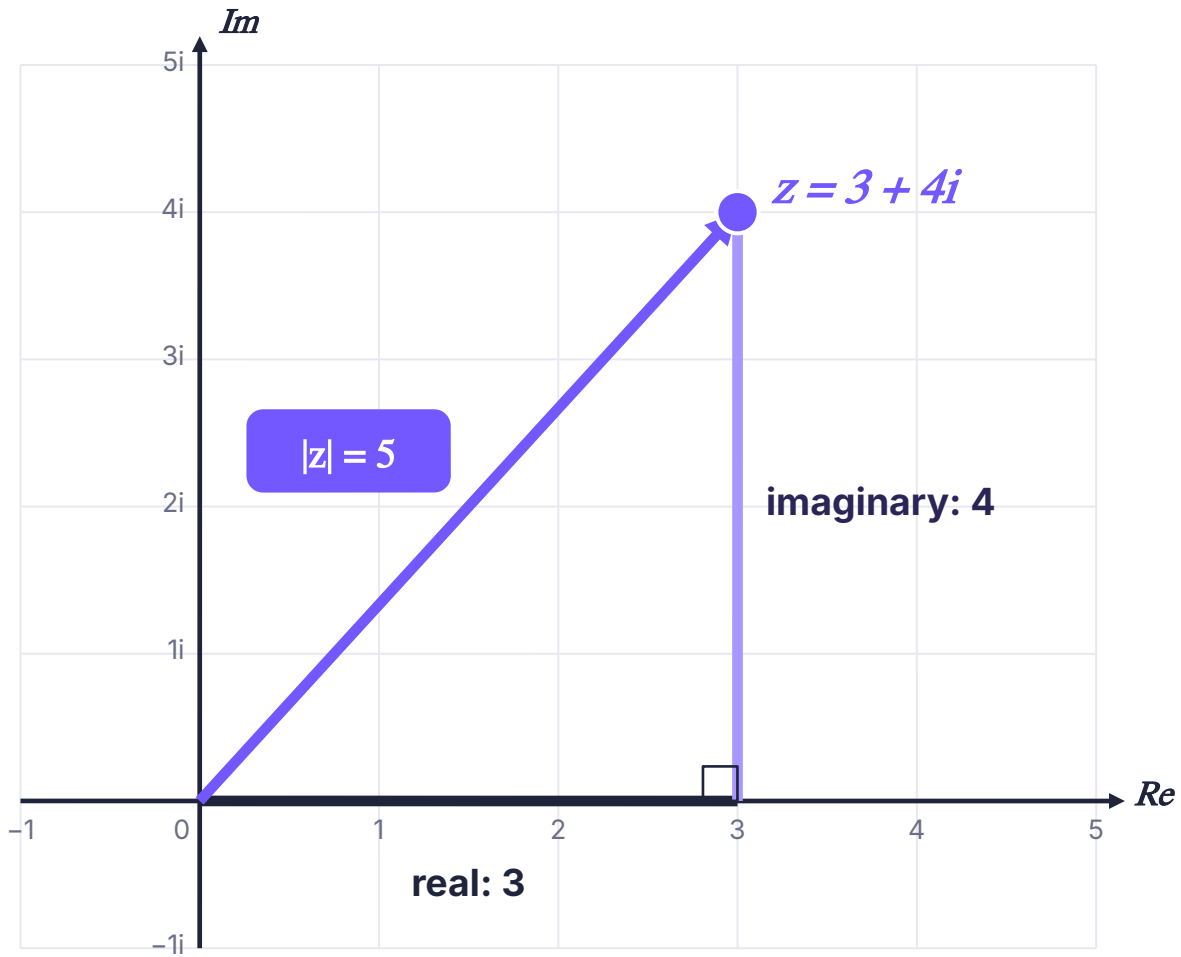
WHERE DOES IT COME FROM?



$$c^2 = a^2 + b^2$$

The Pythagorean theorem! The distance |z| is the long side of a right triangle.

Visualizing Absolute Value



Let's read the triangle:

- **Horizontal side:** real part = 3
- **Vertical side:** imaginary part = 4
- **Long side:** distance $|z|$ = ?

$$|z| = \sqrt{(3^2 + 4^2)}$$
$$= \sqrt{(9 + 16)} = \sqrt{25}$$
$$= 5$$

Calculating Absolute Value: Easy Examples

EXAMPLE 1

$$z = 3 + 4i$$

1	$\lvert z\rvert = \sqrt{(3^2 + 4^2)}$	Square each part
2	$= \sqrt{(9 + 16)}$	Add
3	$= \sqrt{25}$	Take the square root
4	$= 5$	Answer

EXAMPLE 2

$$z = 5 + 12i$$

1	$\lvert z\rvert = \sqrt{(5^2 + 12^2)}$	Square each part
2	$= \sqrt{(25 + 144)}$	Add
3	$= \sqrt{169}$	Take the square root
4	$= 13$	Answer

Properties of Absolute Value

1 $|z| \geq 0$

A distance can never be negative.

e.g. $|3 + 4i| = 5 \geq 0$

2 $|z| = 0$ *only when* $z = 0$

Only the origin is at distance zero.

e.g. $|0 + 0i| = \sqrt{0} = 0$

3 $|\bar{z}| = |z|$

A mirror image is the same distance away.

e.g. $|3 - 4i| = \sqrt{(9 + 16)} = 5$

4 $|z_1 z_2| = |z_1|\,|z_2|$

Multiply the numbers → multiply their sizes.

e.g. $|(3 + 4i)\cdot 2| = |6 + 8i| = 10 = 5 \times 2$

The Important Connection

NUMBER × CONJUGATE = DISTANCE²

$$z \cdot \bar{z} = |z|^2$$

Why is this useful?

- The result is always a real number — no i left.
- A quick way to get $|z|^2$ without a square root.
- The trick for dividing: $1 \div (3 + 4i) = (3 - 4i) \div 25$

LET'S CHECK WITH $z = 3 + 4i$

$$z \cdot \bar{z} = (3 + 4i)(3 - 4i) \quad \text{multiply}$$

$$= 9 - 12i + 12i - 16i^2 \quad \text{expand}$$

$$= 9 - 16i^2 \quad \text{middle terms cancel}$$

$$= 9 + 16 \quad \text{because } i^2 = -1$$

$$= 25$$

$|z|^2 = 5^2 = 25$ ✓ same answer!

Conjugate vs. Absolute Value

	Complex Conjugate $\bar{z}$	Absolute Value $\lvert z \rvert$
What it does	Changes the sign of the imaginary part	Gives the size (magnitude) of z
Result	Another complex number	A non-negative real number
Formula	$\bar{z} = a - bi$	$\lvert z \rvert = \sqrt{(a^2 + b^2)}$
Picture	Reflection across the real axis	Distance from the origin
For $z = 3 + 4i$	$\bar{z} = 3 - 4i$	$\lvert z \rvert = 5$

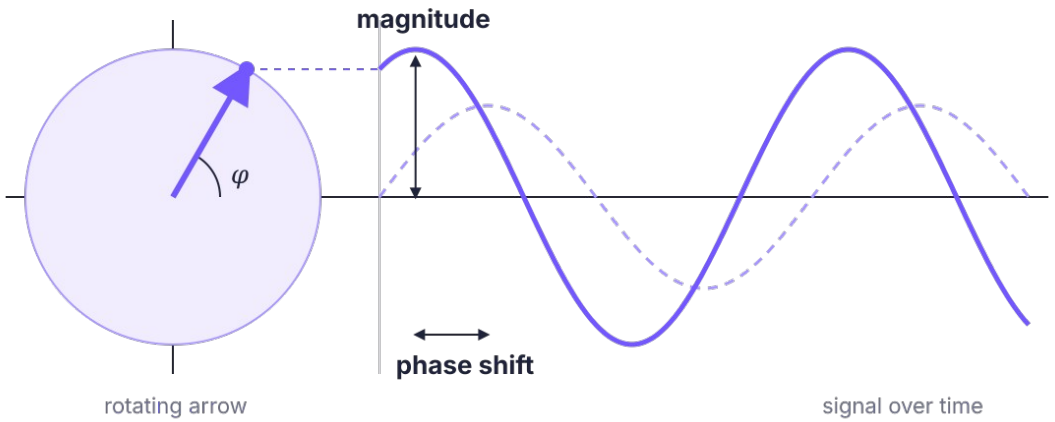
Real-Life Applications

Electrical engineering
AC voltages and currents are written as complex numbers.

Digital signal processing
Filters and audio effects work on complex values.

Communication systems
Wi-Fi and mobile signals carry data in magnitude and phase.

Sound and signal analysis
Each frequency in a sound has a size and a timing.



Arrow length = magnitude (that's $|z|$)
Arrow angle = phase

Key idea: one complex number stores two facts about a signal at once — its magnitude and its phase.

Practice and Quick Quiz

Try each one first — then click to reveal the answer.

Q1 Find the conjugate of

$$z = 4 + 6i$$

$\bar{z} = 4 - 6i$

Q2 Find the absolute value of

$$z = 3 + 4i$$

$|z| = 5$

Q3 Find the conjugate of

$$z = -2 - 5i$$

$\bar{z} = -2 + 5i$

Q4 Calculate $|z|$ if

$$z = 5 + 12i$$

$|z| = 13$

Conclusion & Thank You

- Complex numbers have a real part and an imaginary part: $z = a + bi$.
- A conjugate flips the sign of the imaginary part: $\bar{z} = a - bi$.
- Absolute value is the distance from the origin: $|z| = \sqrt{(a^2 + b^2)}$.
- The identity $z\,\bar{z} = |z|^2$ connects conjugates and absolute value.
- Complex numbers are useful in mathematics, IT, and engineering.

Thank You

Questions are welcome.

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